

Operating Systems Midterm Exam
Fall 2024

Name (First and Last): _____

	Points (15)
Multiple Choice	

Question	Topic	Points (0-5)
1	OS Kernel	
2	Process Model	
3	Scheduler	
4	Virtual Addresses	
5	Page Replacement Algorithms	

	Points (out of 40)
Total	

1. What is the fundamental role of an operating system?
 - ☐ A) To allow direct hardware access by applications
 - ☐ B) To manage hardware resources and provide services to applications
 - ☐ C) To compile and run user programs
 - ☐ D) To perform data backup operations automatically
2. Which component of the Xinu operating system tracks processes that are ready to run?
 - ☐ A) The process table
 - ☐ B) The readylist
 - ☐ C) The I/O buffer
 - ☐ D) The CPU scheduler
3. In a paging system, what happens when a page fault occurs?
 - ☐ A) The process is terminated
 - ☐ B) The OS loads the required page from disk into memory
 - ☐ C) The page table is flushed
 - ☐ D) The CPU resets the program counter
4. How does the Least Recently Used (LRU) page replacement algorithm differ from FIFO?
 - ☐ A) LRU replaces the oldest page in memory
 - ☐ B) LRU replaces the page that has not been used for the longest time
 - ☐ C) LRU is faster but uses more memory than FIFO
 - ☐ D) LRU does not cause page faults
5. Which of the following is a primary advantage of using shared pages in a virtual memory system?
 - ☐ A) It increases the total physical memory available
 - ☐ B) It reduces memory overhead by allowing multiple processes to share common data
 - ☐ C) It prevents all memory leaks in a system
 - ☐ D) It enables faster context switching between processes

6. In the producer-consumer model, what issue arises when the producer adds data to a full buffer?

- ☐ A) Buffer underflow
- ☐ B) Buffer overflow
- ☐ C) Deadlock
- ☐ D) Synchronization

7. What happens during a context switch in the Xinu operating system?

- ☐ A) The CPU halts all processes
- ☐ B) The state of the current process is saved and the next process is loaded
- ☐ C) The memory is reallocated between processes
- ☐ D) The scheduler is reset

8. Which method helps prevent starvation in priority scheduling algorithms?

- ☐ A) First-Come, First-Serve scheduling
- ☐ B) Round Robin scheduling
- ☐ C) Aging mechanism
- ☐ D) Shortest Job First scheduling

9. What role does a Translation Lookaside Buffer (TLB) play in virtual memory?

- ☐ A) It stores the entire page table
- ☐ B) It speeds up the translation of virtual addresses to physical addresses
- ☐ C) It handles page faults
- ☐ D) It prevents memory fragmentation

10. Which of the following is true about memory-mapped I/O?

- ☐ A) It reduces overhead by allowing I/O devices to be accessed as if they were memory locations
- ☐ B) It separates the I/O and memory spaces entirely
- ☐ C) It prevents direct access to I/O devices by the CPU
- ☐ D) It ensures that each I/O operation is handled synchronously

11. How does the Xinu operating system keep track of free memory?
- ☐ A) It uses a bitmap to represent free and allocated memory blocks
 - ☐ B) It maintains a free list that links together blocks of free memory
 - ☐ C) It stores free memory addresses in a table sorted by size
 - ☐ D) It uses paging to manage memory allocation and free space
12. How does the operating system resolve a page fault?
- ☐ A) By terminating the process that caused the page fault
 - ☐ B) By loading the required page from swap space into physical memory
 - ☐ C) By increasing the size of the TLB
 - ☐ D) By resetting the page table
13. What is the primary challenge of managing free memory using a linked list in Xinu?
- ☐ A) It causes internal fragmentation when blocks are split
 - ☐ B) The free list can become unsorted, making memory allocation inefficient
 - ☐ C) It results in external fragmentation as free memory gets split into smaller blocks
 - ☐ D) It requires large amounts of memory to store the free list
14. What is the primary advantage of using semaphores in the producer-consumer model?
- ☐ A) They prevent data corruption by synchronizing access to shared buffers
 - ☐ B) They increase the speed of buffer operations
 - ☐ C) They remove the need for memory management in shared systems
 - ☐ D) They allow the consumer to produce data as well
15. What type of memory allocation strategy does Xinu primarily use to allocate free memory?
- ☐ A) Best Fit
 - ☐ B) Worst Fit
 - ☐ C) First Fit
 - ☐ D) Next Fit

1. OS Kernel

1. What is the purpose of system calls in the context of modern operating systems?
2. Why does kernel space exist?

2. Process Model

1. Explain the role of the process table in managing processes in Xinu.
2. Describe how the scheduling algorithm in Xinu interacts with the process table and the ready queue to manage process execution.
3. In your answer, discuss the data structures used for the process table and ready queue, and how context switching is implemented.

3. Scheduler

Your operating system uses a Round Robin scheduler with a time quantum of 100 ms. What happens if a process's I/O wait is shorter than the quantum? How can this behavior be optimized to reduce context switches?

4. Virtual Addresses

What happens during a context switch between processes in the Xinu operating system, and how does the system maintain process isolation?

5. Page Replacement Algorithm

Scenario:

You are a systems engineer working on an embedded system that currently uses the First-In-First-Out (FIFO) page replacement algorithm. The system has 1 GB of memory and handles small, predictable workloads. You are tasked with switching the system to the Least Recently Used (LRU) algorithm to handle new, unpredictable workloads.

Question:

Explain how the transition from FIFO to LRU will increase the memory overhead of the OS. What additional data structures (such as counters, linked lists, or stacks) does LRU require compared to FIFO? Discuss how these structures impact both memory usage and system performance.